Model Specification and Statistical Control

November 7, 2022

As You Come In...

Pick up a worksheet. If you get it all right, $+2$ extra credit. If not but you tried, $+1$.

Outline

1. Levels of measurement and regression
2. When you need to control
3. Data transformations
4. Interaction terms

Rule 1: Know the levels of measurement of your variables

Levels of Measurement and Independent Variables

- Interval: Include in regression as single, interval-level variable.
  - Interpret as a 1 unit change in X is associated with a β unit change in Y, on average, holding all other variables constant

Levels of Measurement and Independent Variables

- Interval: Include in regression as single, interval-level variable.
  - Interpret as a 1 unit change in X is associated with a β unit change in Y, on average, holding all other variables constant
- Nominal: Include as a series of dichotomous variables
  - If variable is binary (only two categories), just include the one variable
  - if variable has multiple categories (e.g., region), include a binary variable for each category and leave out one as baseline
  - Interpretation compares each category to baseline
  - R does this for you: factor()

	Dependent variable:
	% Dems in Leg.
Southern State	−2.627
	(4.578)
Constant	56.350*
	(2.616)
Observations	49
R^2	0.007
Adjusted R^2	−0.014
Residual Std. Error	15.029
Note:	*p<0.05

Table: Linear regression with categorical independent variables.

	Dependent variable:	
	% Leg. Dems.	
Western State	−1.067	−16.180*
	(5.180)	(6.015)
Midwestern State	−3.223	−18.336*
	(5.433)	(6.235)
Northeastern State	15.113*	
	(5.780)	
Southern State		−15.113*
		(5.780)
Constant	53.722*	68.836*
	(3.468)	(4.624)
Observations	49	49
R^2	0.190	0.190
Adjusted R^2	0.136	0.136
Note:	*p<0.05	

Levels of Measurement and Independent Variables

- Interval: Include in regression as single, interval-level variable.
  - Interpret as a 1 unit change in X is associated with a β unit change in Y, on average, holding all other variables constant
- Nominal: Include as a series of dichotomous variables
  - If variable is binary (only two categories), just include the one variable
  - if variable has multiple categories (e.g., region), include a binary variable for each category and leave out one as baseline
  - Interpretation compares each category to baseline
  - R does this for you: factor()
- Ordinal
  - If distances between categories are same, treat as interval
  - If distances between categories are not same, treat as categorical

Rule 2: Model Specification (e.g., which variables we include) should be driven by <u>THEORY</u>.

Controlling Won't Always Save You

"Based on the behavioral priming literature in psychology, which shows that activating one mental construct increases the tendency of people to engage in mentally related behaviors, one may conjecture that activating '**oddness**,' may lead people to act in less traditional ways, e.g., seeking information from non-traditional sources. I used data from the GSS and examined if respondents who were randomly assigned an odd respondent ID (1,3,5...) were more likely to report reading horoscopes."

Controlling Won't Always Save You

Table 1. Impact of having an odd number ID on reading the horoscope.

	Baseline
Respondent's id is an ODD number	**0.111*** **
	(0.038)
Age	
Female (1/0)	
Confidence in belief in God *(1-don't-believe to 6-Know God Exists)*	
Happiness *(1-Not too happy, 2 -Somewhat, 3-Happy)*	
Taken college science class (1-yes, 0-No)	
Political affiliation (k=7)	NO
Observations	689
R-squared	0.012

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Controlling Won't Always Save You

Table 1. Impact of having an odd number ID on reading the horoscope.

	Baseline	Add Demographics	Add Personality	Add College Science
Respondent's id is an ODD number	**0.111*** **	**0.117*** **	**0.107*** **	**0.106*** **
	(0.038)	*(0.037)*	*(0.037)*	*(0.037)*
Age		-0.001	-0.001	-0.001
		(0.001)	*(0.001)*	*(0.001)*
Female (1/0)		0.248***	0.251***	0.252***
		(0.037)	*(0.038)*	*(0.038)*
Confidence in belief in God			-0.009	-0.01
(1-don't-believe to 6-Know God Exists)			*(0.013)*	*(0.014)*
Happiness			0.109***	0.107***
(1-Not too happy, 2 -Somewhat, 3-Happy)			*(0.028)*	*(0.028)*
Taken college science class (1-yes, 0-No)				-0.021
				(0.042)
Political affiliation (k=7)	NO	NO	YES	YES
Observations	689	687	666	666
R-squared	0.012	0.076	0.111	0.111

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Controlling Won't Always Save You

People are about 11 percentage points more likely to read the horoscope when they are randomly assigned an odd number by the GSS. Moreover, this estimate barely changes across alternative specifications that include more and more covariates, despite the notable increase in R^2.

What if We Don't Control When We Should?

Omitted Variable Bias- Very, Very Bad Thing

Let's say our <u>true</u>, population equation is:

$$Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon_i \tag{1}$$

But we estimate our model as:

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}_1 X_1 \tag{2}$$

- Now, $\hat{\beta}_1 \neq \beta_1$; instead, $\hat{beta}_1$ is β_1 plus a bunch of stuff that X_2 and X_3 would otherwise be accounting for...
- Especially bad if magnitude of βs are large
- Usually we don't even know which direction the bias goes

When Do We Need to Control?

TABLE 10.8 Some Three-Variable Relationships

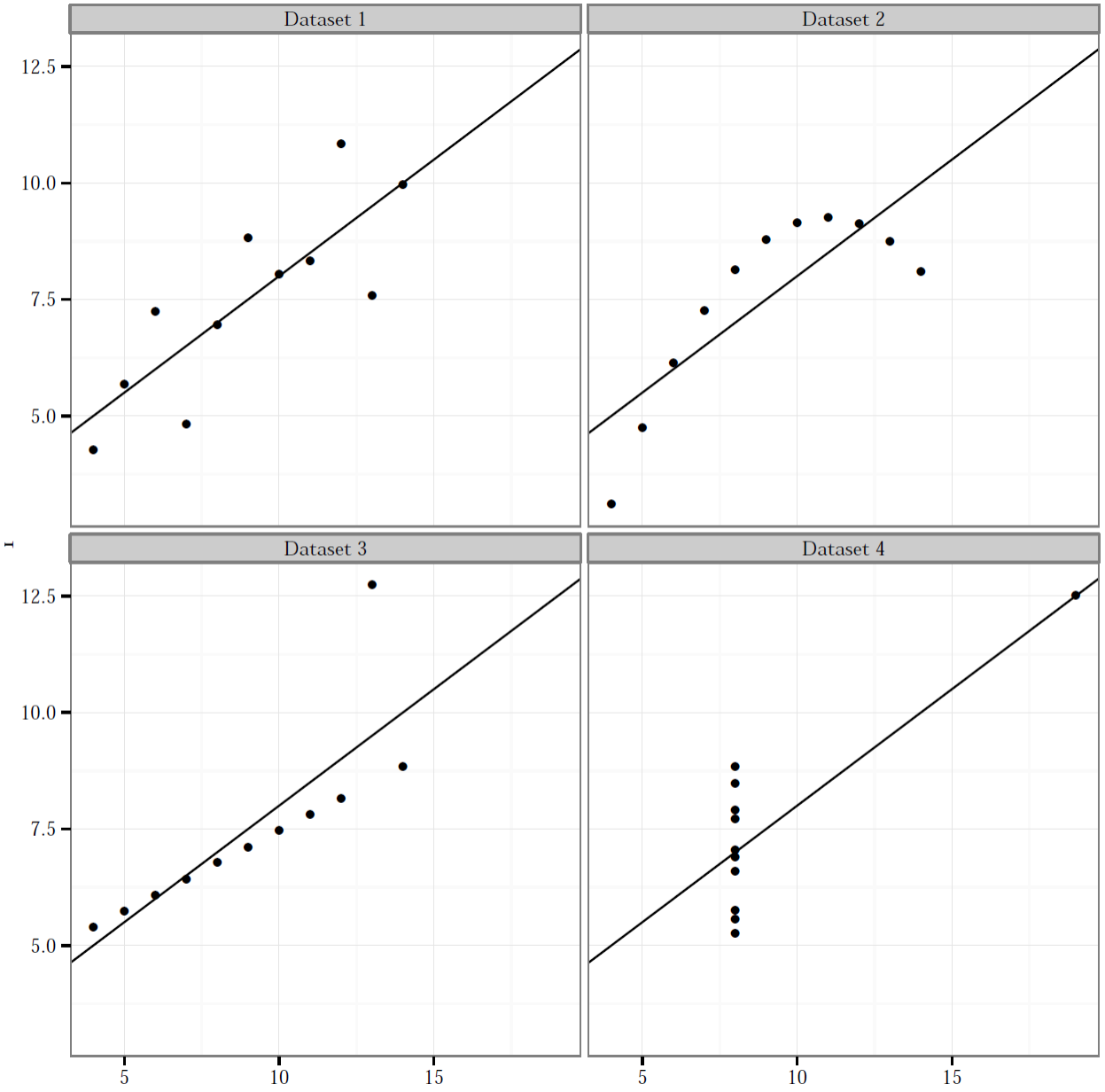
	Graph	Name of Relationship	What Happens After Controlling for X_2
YES	$X_2 \nearrow X_1$, $X_2 \searrow Y$	Spurious X_1–Y association	Association between X_1 and Y disappears.
NO	$X_1 \longrightarrow X_2 \longrightarrow Y$	Chain relationship; X_2 intervenes; X_1 indirectly causes Y	Association between X_1 and Y disappears.
YES	$X_2 \downarrow$, $X_1 \longrightarrow Y$	Interaction	Association between X_1 and Y varies according to level of X_2.
NO	$X_2 \searrow Y$, $X_1 \nearrow Y$	Multiple causes	Association between X_1 and Y does not change.
YES	$X_1 \longrightarrow Y$, $X_1 \searrow X_2 \nearrow Y$	Both direct and indirect effects of X_1 on Y	Association between X_1 and Y changes, but does not disappear.

Rule 3: <u>Always</u> plot your data first

Reason 1: Anscombe's Quartet

Statistic	Value
$\bar{x}$	9
$\bar{y}$	7.50
r	0.82
Regression	$y = 3.00 + 0.50x$

Reason 1: Anscombe's Quartet



Reason 2: Data Transformations

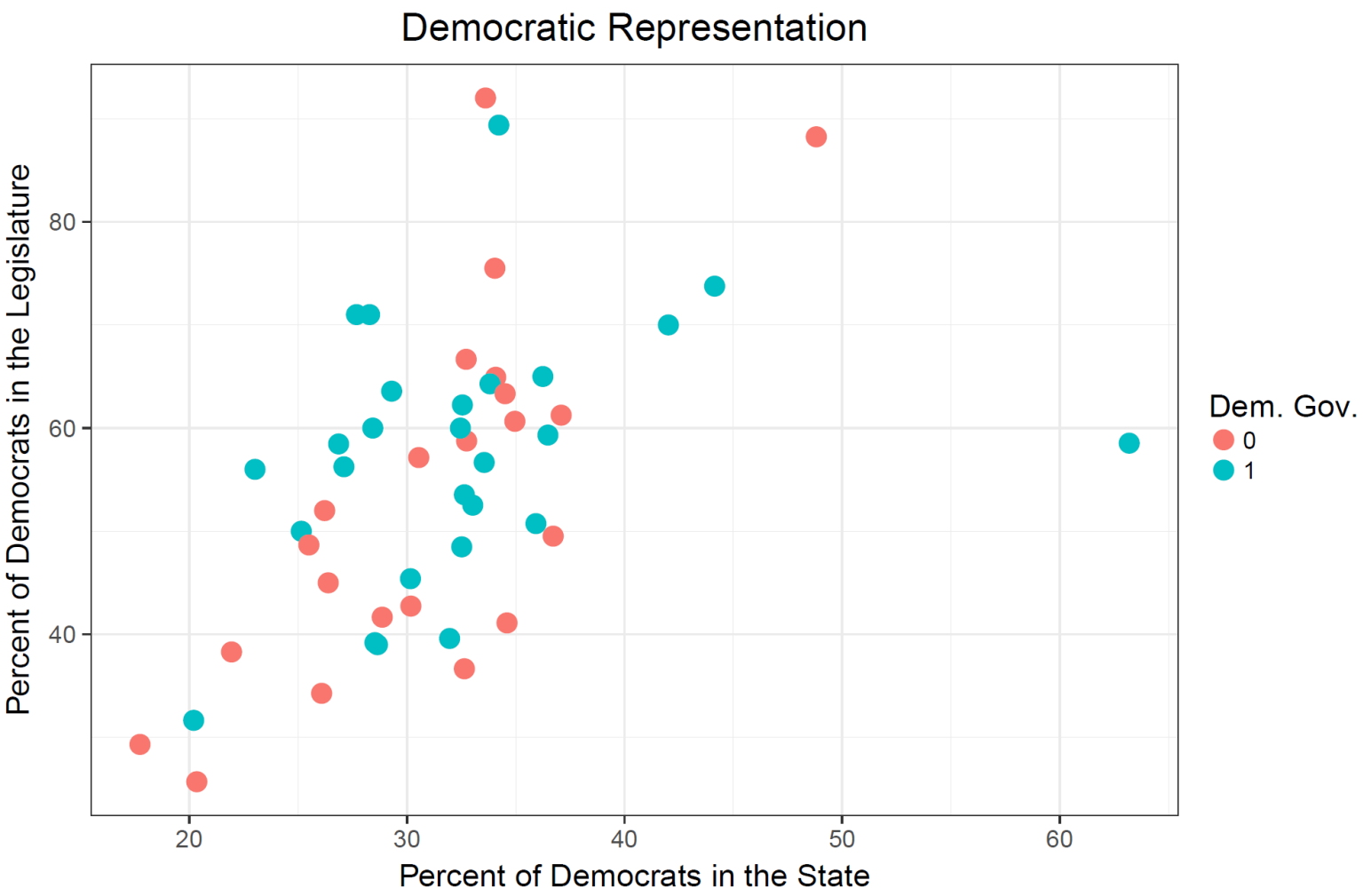
Linear Regression requires a linear relationship between X and Y.

Sometimes there may not appear to be a relationship until you reshape data to fit shape

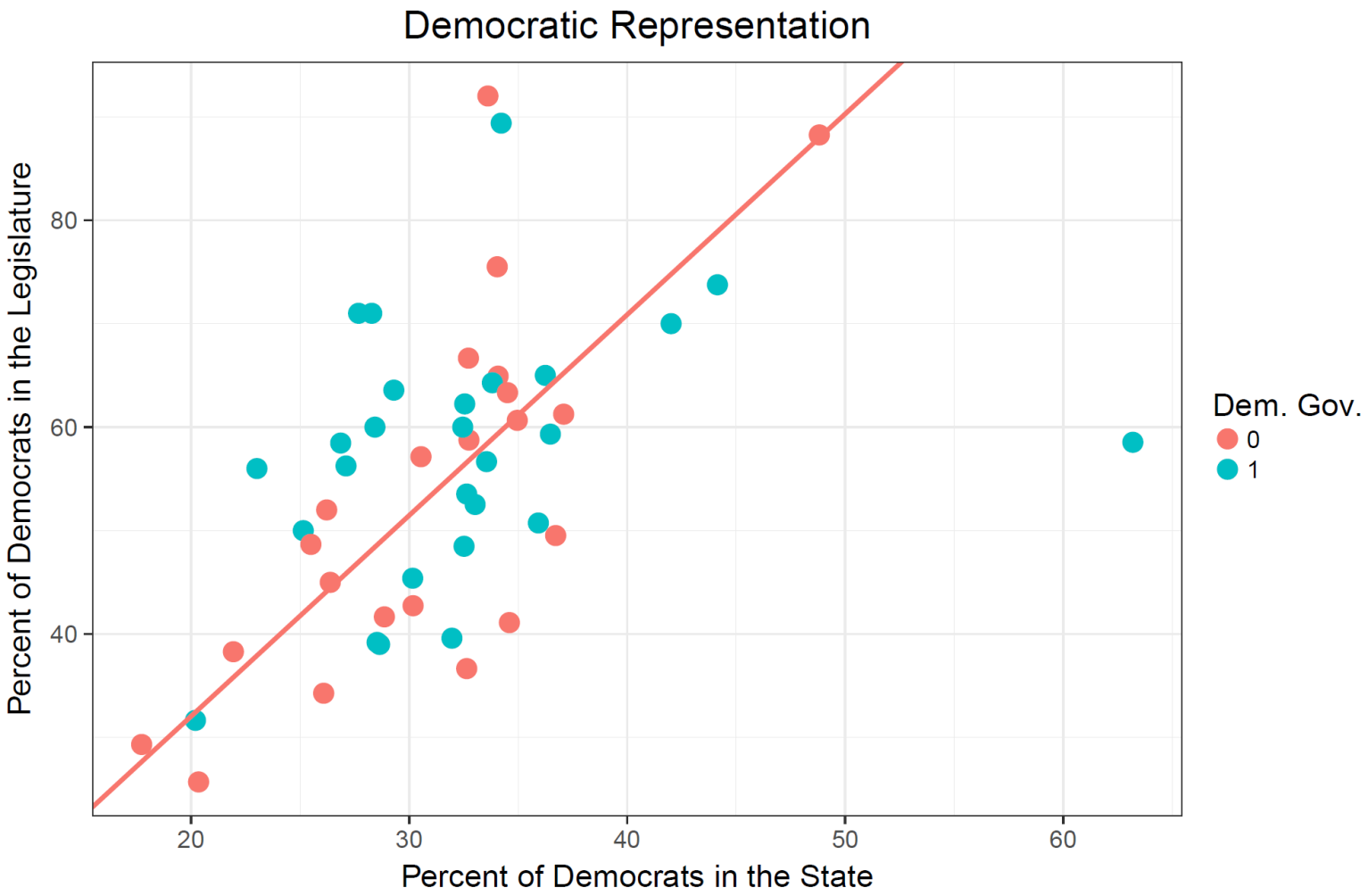
- Quadratic (U-shaped): include X AND X^2 - direction of first β is how relationship starts, direction of second *beta* is how relationship ends...
- Logarithmic: replace X with $log(X)$ to account for skew in X or when expect effects to dissipate

Rule 4: Properly specify and interpret <u>interaction terms</u>

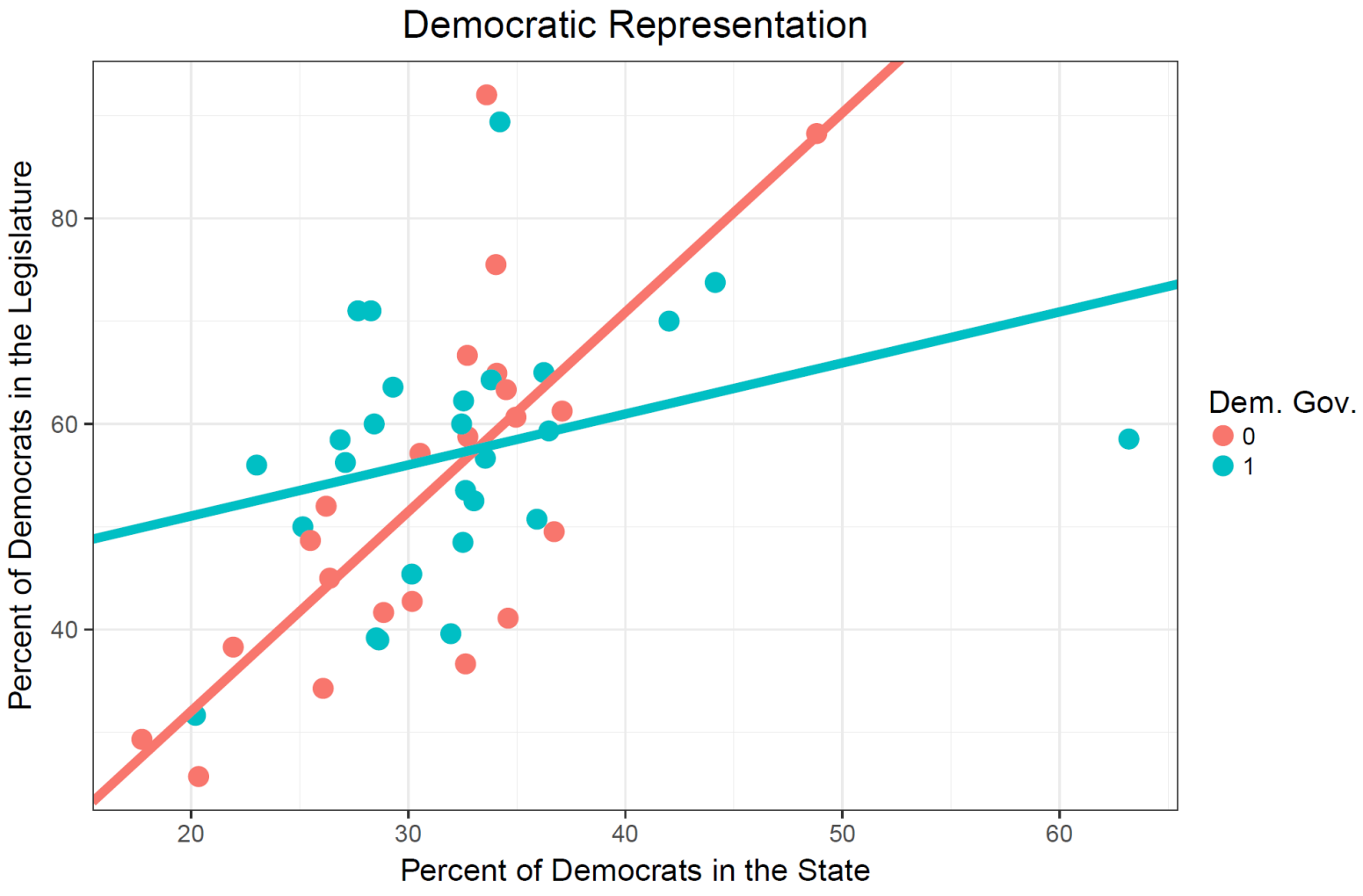
Interaction Example



Interaction Example



Interaction Example

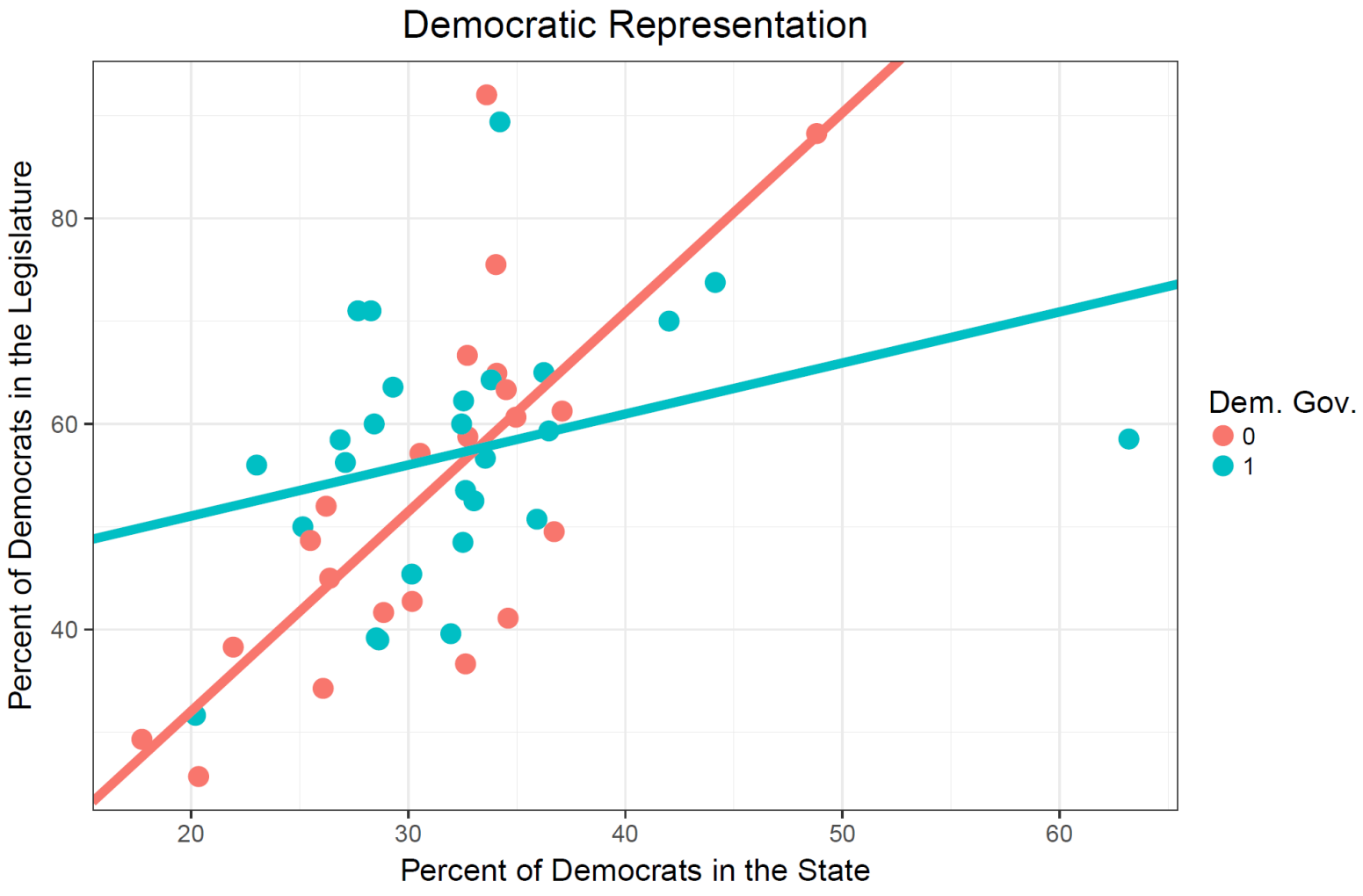


Multiplicative Interactions

If the effect of X_1 on Y varies based on the level of X_2, we include a <u>multiplicative interaction</u> between X_1 and X_2.

$$Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + ... + \epsilon_i \tag{3}$$

Interaction Example



Interaction Example

$$
\begin{aligned}
\text{\% Leg Dems} = &-6.7 \\
&+47.9 \times \text{Dem. Gov.} \\
&+1.9 \times \text{\% Dem.} \\
&-1.4 \times \text{Dem. Gov.} \times \text{\% Dem.}
\end{aligned}
$$

Interaction Example

```
lm(formula = legDems ~ democrat * govparty_c, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-20.009  -8.194   1.623   5.876  33.437

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)           -6.733     12.452  -0.541  0.59134
democrat               1.943      0.394   4.931 1.16e-05 ***
govparty_c            47.861     15.884   3.013  0.00424 **
democrat:govparty_c   -1.447      0.492  -2.941  0.00515 **
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

Residual standard error: 12.07 on 45 degrees of freedom
Multiple R-squared:  0.387,Adjusted R-squared:  0.3462
F-statistic: 9.471 on 3 and 45 DF,  p-value: 5.768e-05
```

Interpreting Interactions

$$Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + ... + \epsilon_i \tag{4}$$

- $\hat{\beta}_1$ = effect of X_1 on Y when $X_2 = 0$
- $\hat{\beta}_2$ = effect of X_2 on Y when $X_1 = 0$
- $\hat{\beta}_3$ = effect of $X_1 \times X_2$ on Y...

Calculating Interactions

Given the following model:

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_1 X_2 \tag{5}$$

The effect of X_1 is:

$$\hat{\beta}_1 + \hat{\beta}_3 X_2$$

The effect of X_2 is:

$$\hat{\beta}_2 + \hat{\beta}_3 X_1$$

Specifying Interactions in Sum

- Include interactive effects when you expect the effect of one X on Y to vary according to the levels of another X
- You must <u>always</u> include X_1 and X_2 in the model
  - Another easier, approximately correct approach is to subset on nominal variables and run separate regressions
- Remember you must interpret $\hat{\beta}_1$ and $\hat{\beta}_2$ very carefully!

Rules for Model Specification

1. Know your levels of measurement!
2. Model specification based on theory
3. Always plot data before regressing
4. Specify and interpret interaction terms properly: include constituent terms or subset on categorical variable